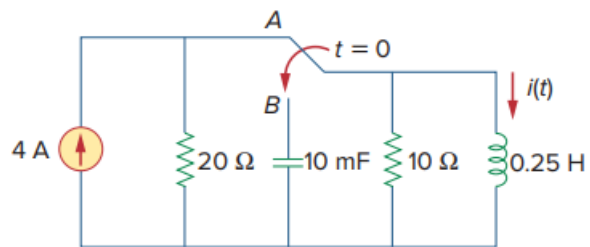


Problem

The switch in Fig. 8.77 moves from position A to position B at $t = 0$ (please note that the switch must connect to point B before it breaks the connection at A, a make-before-break switch). Determine $i(t)$ for $t > 0$.

Figure 8.77:



Step-by-step solution

Step 1 of 8

Draw the circuit diagram.

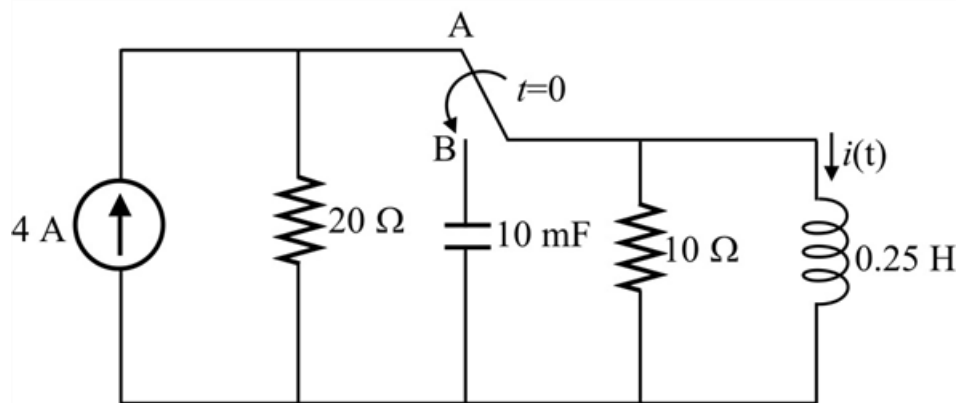


Figure 1

Step 2 of 8

From the movement when $t = 0^-$ the switch in the circuit is connected to position A and the circuit is reached steady state, it means that the circuit has reached dc steady state at $t = 0^-$. The inductor acts like a short circuit, while the capacitor acts like an open circuit. Hence the circuit becomes as shown in Figure 2.

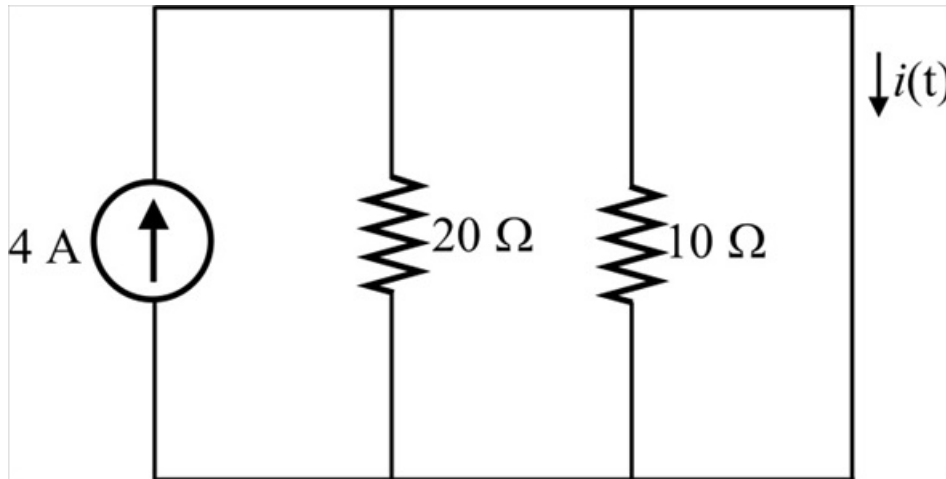


Figure 2

Step 3 of 8

As inductor is short circuited current flows through short only rather than resistance

Hence initial current through inductor is:

$$i(0^-) = 4 \text{ A}$$

At $t > 0$, switch is thrown to position B, that is source is disconnected from the circuit. Now the circuit will be like as below in Figure 3.

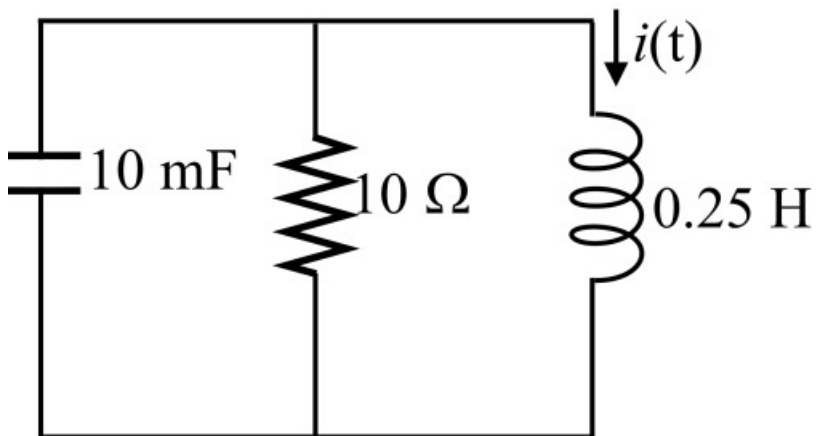


Figure 3

Step 4 of 8

From Figure 3, it is a parallel RLC circuit. Calculate the response.

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Here $\alpha = \frac{1}{2RC}$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The roots are calculated as follows:

$$\begin{aligned}\alpha &= \frac{1}{2RC} \\ &= \frac{1}{(2)(10)(10 \times 10^{-3})} \\ &= \frac{1}{0.2} \\ &= 5\end{aligned}$$

Step 5 of 8

Calculate the resonant frequency.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(0.25)(10 \times 10^{-3})}} \\ &= \frac{1}{\sqrt{(2.5 \times 10^{-3})}} \\ &= 20\end{aligned}$$

The roots are calculated as follows:

$$\begin{aligned}s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \\ &= -5 \pm \sqrt{5^2 - 20^2} \\ &= -5 \pm j19.365\end{aligned}$$

As $\alpha < \omega_0$, the response is underdamped.

Hence the voltage is,

$$i(t) = e^{-5t} (A_1 \cos 19.365t + A_2 \sin 19.365t) \dots\dots (1)$$

Substitute 0 for t .

$$\begin{aligned}i(0) &= A_1 + 0 \\ &= A_1\end{aligned}$$

Substitute 4 for A_1 .

$$A_1 = 4$$

Step 6 of 8

Calculate the voltage across inductor.

$$v(t) = L \frac{di}{dt}$$

$$\frac{di(0)}{dt} = \frac{v(0)}{L}$$

As inductor is short circuited its voltage is zero.

$$\frac{di(0)}{dt} = \frac{0}{0.25}$$

$$= 0 \text{ A/s}$$

Step 7 of 8

Differentiate equation (1).

$$\frac{di(t)}{dt} = e^{-st} \begin{pmatrix} -5A_1 \cos 19.365t - 5A_2 \sin 19.365t \\ -19.365A_1 \sin 19.365t + 19.365A_2 \cos 19.365t \end{pmatrix}$$

Put 0 for t .

$$\frac{di(0)}{dt} = e^{-s(0)} \begin{pmatrix} -5A_1 \cos 19.365(0) - 5A_2 \sin 19.365(0) \\ -19.365A_1 \sin 19.365(0) + 19.365A_2 \cos 19.365(0) \end{pmatrix}$$

$$= -5A_1 - 0 - 0 + 19.365A_2$$

$$= -5A_1 + 19.365A_2$$

Substitute 0 for $\frac{di(0)}{dt}$.

$$-5A_1 + 19.365A_2 = 0$$

$$19.365A_2 = 5A_1$$

Substitute 4 for A_1 .

$$19.365A_2 = 5(4)$$

$$A_2 = \frac{20}{19.365}$$

$$= 1.033$$

Step 8 of 8

Substitute 4 for A_1 and 1.033 for A_2 in $i(t) = e^{-st} (A_1 \cos 19.365t + A_2 \sin 19.365t)$.

$$i(t) = e^{-st} (4 \cos 19.365t + 1.033 \sin 19.365t)$$

Therefore, the current $i(t)$ for $t > 0$ is $e^{-st} (4 \cos 19.365t + 1.033 \sin 19.365t)$.